DATA SCIENCE PROJECT

Bike Sharing Demand Prediction

Objective: Predict the hourly number of bike rentals using temporal, calendar, and weather-related features

🎤 **Presentation:** Heike Thöricht

📍 **Bremen,** 23. August 2026

Agenda

The Challenge

Why demand forecasting matters

Our Approach

Data-driven solution

Model Performance

Proven accuracy

Key Insights

What drives demand

Implementation & Next Steps

How to use it

Executive Summary — Ready for Go-Live

Prediction Accuracy

95.7% of demand variance explained ($R^2 = 0.957$) — exceptional accuracy for confident planning

Forecast Error

±29 bikes on average (MAE = 29.1) — very tight margin for precise logistics planning

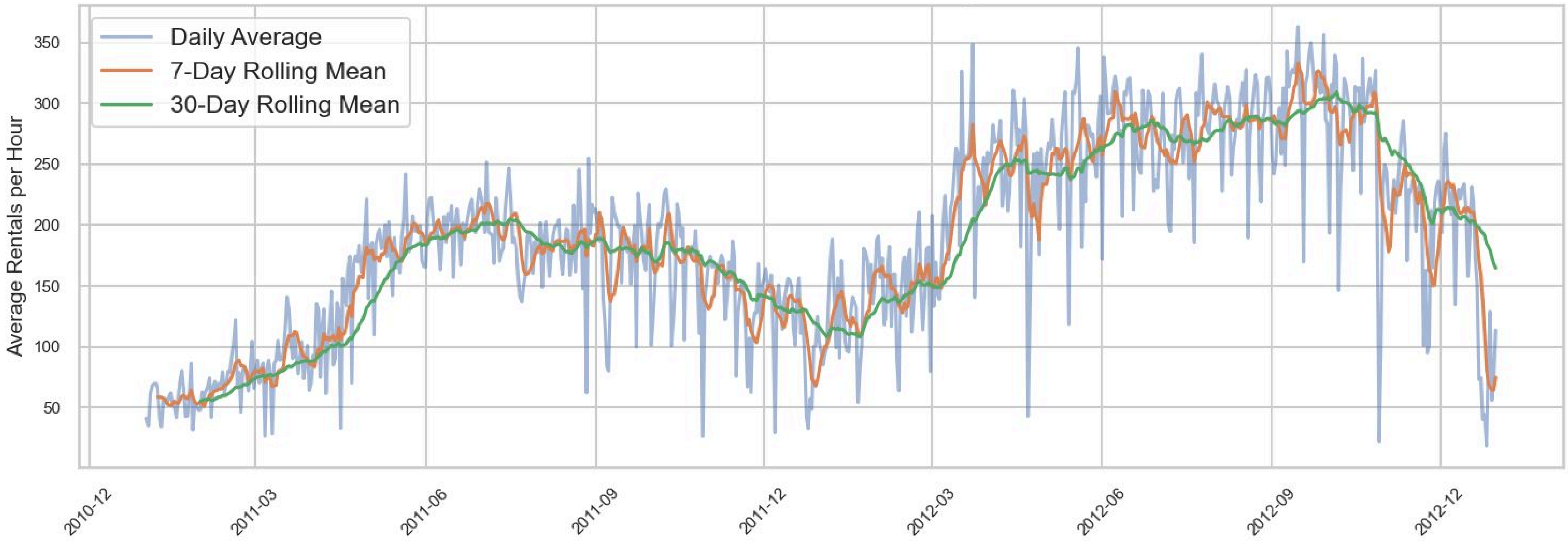
Key Driver

Historical demand patterns (lag_1: 36%) and time-of-day (18.1%) drive predictions — highly actionable

The model achieves exceptional accuracy using historical demand patterns and temporal features — production-ready to support your planning processes.

EXPLORATORY ANALYSIS

The Rise of Bike Sharing: Emerging Business Opportunities



The Challenge: Why Demand Forecasting Matters

Our Data Foundation

Real-world hourly rental data from the Capital Bikeshare system — a proven operational dataset that mirrors your planning environment.

- **17,379** hourly observations from real bike-sharing operations
- **2 years** of historical data (2011–2012)
- Complete data quality — no missing values

The Business Problem

Without reliable demand forecasts, your planning division faces avoidable inefficiencies and operational risk.

- Unpredictable demand creates **logistics and staffing challenges**
- Core question: **How many bikes are needed each hour?**
- Key insight: **Historical demand patterns (36%)** + **time-of-day (18%)** are the strongest predictors
- Goal: Accurate hourly forecasts based on **demand momentum and temporal patterns**

Data source: Fanaee-T, H. (2013). Bike Sharing [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5W894.

EXPLORATORY ANALYSIS

Key Insight: Demand Patterns Drive Planning

🕗 Commuter Peaks

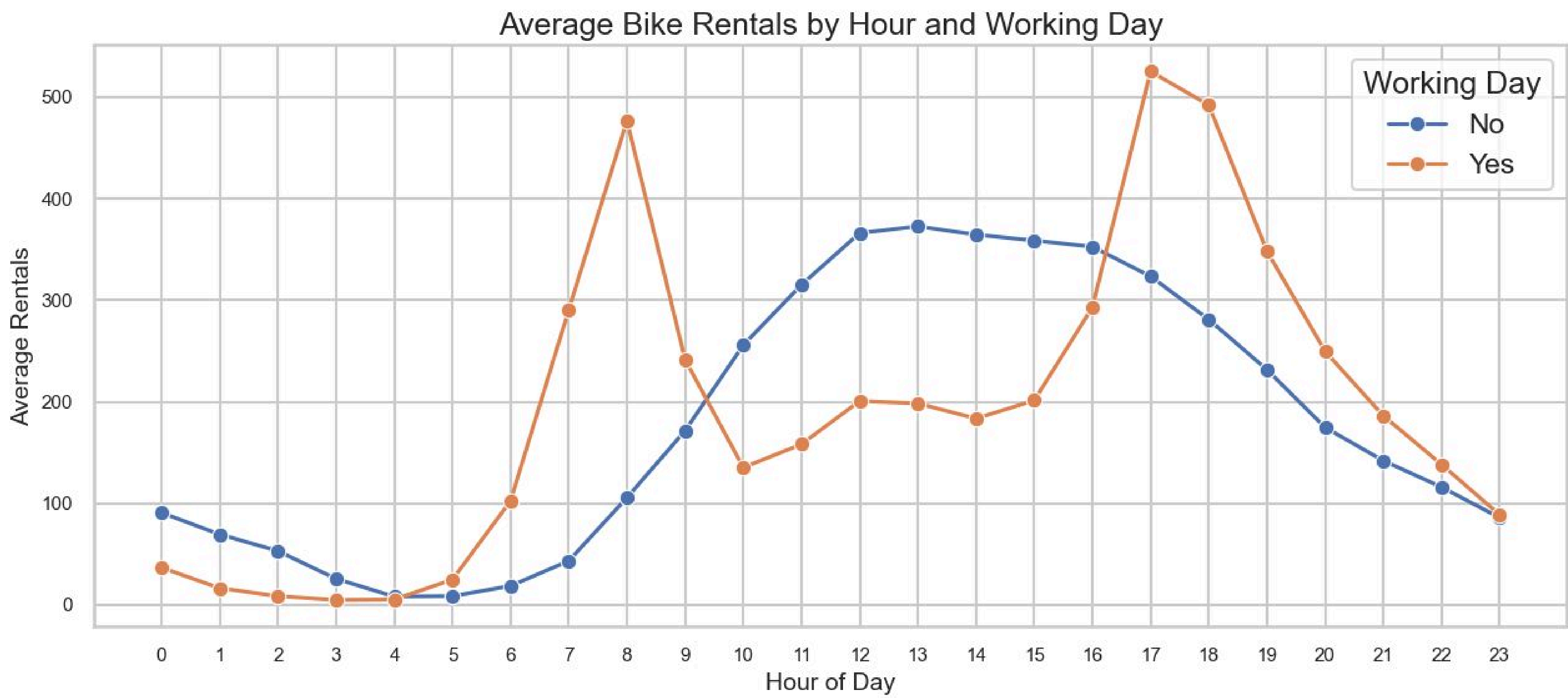
Working days show demand spikes at **8 AM** and **5–6 PM** — align bike availability with commute peaks.

🌤️ Leisure Patterns

Non-working days show midday peaks — different staffing and bike distribution needed

📊 Actionable

Hour-of-day is the **strongest predictor** — enables precise hourly planning



These patterns are consistent and predictable — your planning can rely on them.

MODEL EVALUATION

Model Performance: Proven Accuracy

Comparing our top three models across key performance metrics — XGBoost delivers **exceptional accuracy** with an R^2 of 0.957, making it the clear choice for production deployment.

Model	Feature Set	R^2	MAE	RMSE
Linear Regression	V01	0.9033	47.99	68.56
Random Forest	V01	0.9547	28.99	46.91
🏆 **XGBoost**	**V01**	**0.9575**	**29.12**	**45.48**

🏆 Winner: XGBoost

95.7% accuracy — explains nearly 96 out of 100 demand variations, delivering exceptional performance for production deployment

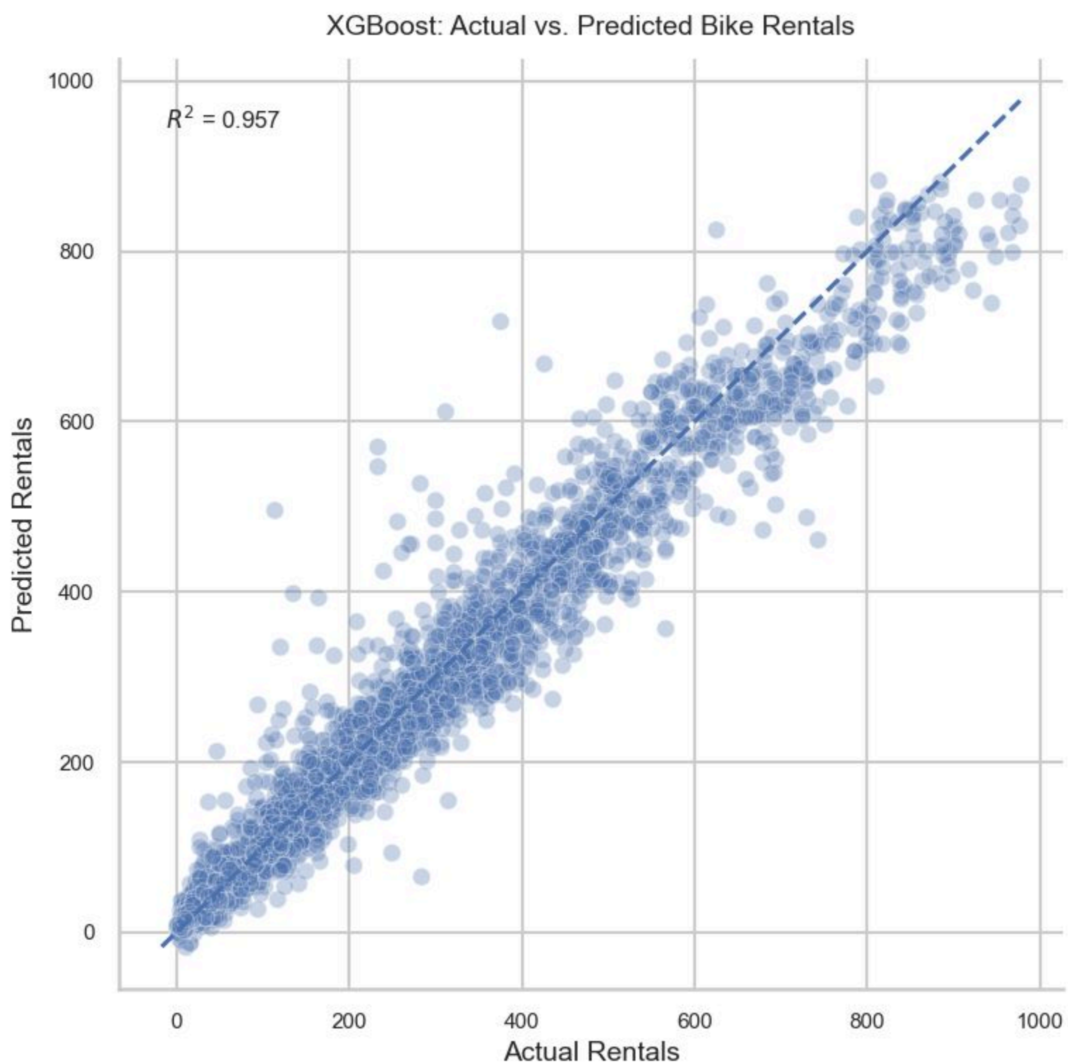
📊 Forecast Error: ±29 Bikes

Average prediction error of just **29 rentals** — a very tight margin for operational planning and fleet distribution

✅ Production-Ready

Tested on **real time-series data** — proven to generalise beyond training conditions and work in live scenarios with exceptional accuracy

Validation: Model Works in Real Scenarios



✅ Accurate Predictions

Model predictions follow actual demand closely — **reliable for planning.**

⚠️ Peak Hour Challenges

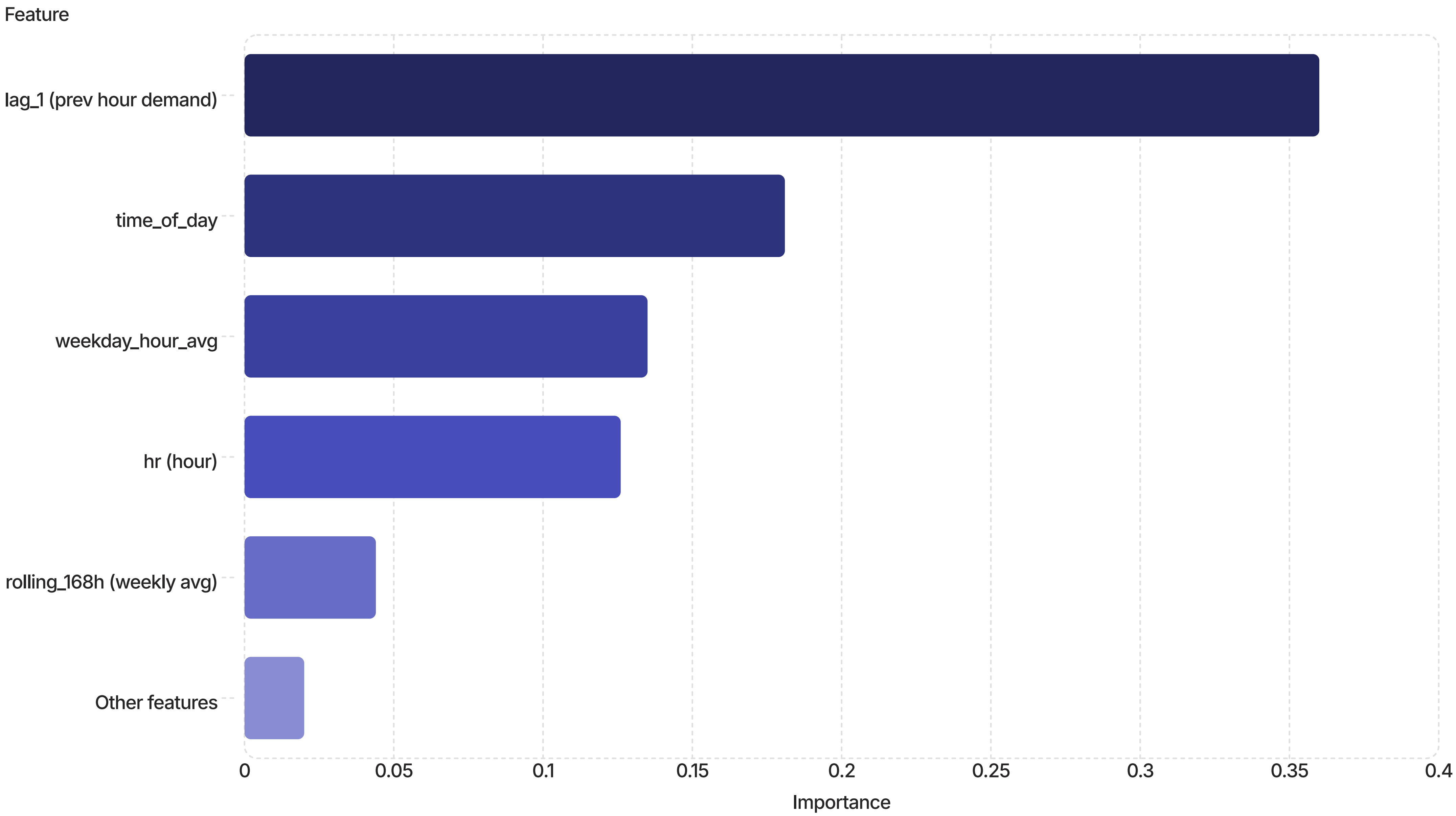
Errors slightly larger during peak hours — expected, but still **within acceptable range**.

🎯 Ready for Operations

Strong overall performance across all demand levels — **suitable for production use**.

The model is validated and ready to support your daily planning operations.

What Drives Demand: The Key Factors



📊 Demand Momentum Dominates

Previous hour's demand (lag_1: 36%) is the strongest predictor — **demand patterns are highly persistent**.

⏰ Time Patterns Matter

Time-of-day (18.1%) and hour-of-day (12.6%) together account for **30.7%** — temporal patterns are critical.

💡 Actionable

Focus on **recent demand trends + hourly patterns** — these drive 67% of predictions. Weather has minimal impact (<3%).

> The model reveals the true drivers: demand momentum (36%) + temporal patterns (31%) together explain **67% of predictions.** Recent history is your best predictor.

Implementation: How to Use the Model

📊 Forecast Refresh & Retraining

- **Daily:** Generate hourly forecasts each evening
- **Weekly:** Retrain model every Monday
- **Why:** lag_1 (36%) needs fresh data; weekday_hour_avg (13.5%) needs stable patterns

Owner: Data/Analytics

🚀 Go-Live Readiness

- Production-ready
- **Requires:** API integration, data pipeline, dashboard
- **Timeline:** 2–3 weeks to deployment

Owner: IT

📈 Success Metrics

- **Target:** MAE < 50 bikes
- **Monitor:** Weekly forecast accuracy
- **Retrain:** Quarterly or when drift detected

Owner: Analytics

The model is ready. Assign owners, build the pipeline, and start forecasting from day one.

Questions & Next Steps

Ready to deploy. Let's discuss implementation timeline and resource allocation.

Heike Thöricht

Action Items

Review forecast dashboard prototype — *this week*

Confirm API requirements with IT — *next week*

Schedule go-live meeting — *within 2 weeks*